

Python Coding Questions & Answers

SET-1	
1)	Write a program that finds all elements appearing in both lists (duplicates allowed only once). And upload add the code to Git repository <i>Example:</i> <pre>List1 = [1, 2, 3, 4] List2 = [3, 4, 5, 6] Output → [3, 4]</pre> <pre>def find_common_elements(list1, list2): common = [] for item in list1: if item in list2 and item not in common: common.append(item) return common</pre> # Example <pre>list1 = [1, 2, 3, 4] list2 = [3, 4, 5, 6]</pre> <pre>result = find_common_elements(list1, list2) print(result) # Output: [3, 4]</pre>
2)	Given a sentence, write a program that counts how many times each word appears. And upload add the code to Git repository <i>Example:</i> <pre>Input: "the cat and the hat" Output: {'the': 2, 'cat': 1, 'and': 1, 'hat': 1}</pre> <pre>def word_frequency(sentence): # Convert to lowercase sentence = sentence.lower() # Split into words words = sentence.split() # Dictionary to store word counts freq = {} for word in words: if word not in freq: freq[word] = 1 else: freq[word] += 1 return freq</pre> # Example

	<pre>sentence = "the cat and the hat" result = word_frequency(sentence) print(result) # Output: {'the': 2, 'cat': 1, 'and': 1, 'hat': 1}</pre>
SET-2	
3)	Given a 2D list (matrix), write Python code that prints its transpose. And upload add the code to Git repository <i>Example:</i> Input: [[1, 2], [3, 4], [5, 6]] Output: [[1, 3, 5], [2, 4, 6]]
4)	Write a program that reverses the order of words in a string. And upload add the code to Git repository <i>Example:</i> "hello world python" → "python world hello" <pre>def transpose_matrix(matrix): # Use list comprehension to transpose transposed = [[row[i] for row in matrix] for i in range(len(matrix[0]))] return transposed # Example matrix = [[1, 2], [3, 4], [5, 6]] result = transpose_matrix(matrix) print(result) # Output: [[1, 3, 5], [2, 4, 6]]</pre>
SET-3	
5)	Write a Python program that removes duplicates from a list without using set(), while preserving the original order. And upload add the code to Git repository <i>Input:</i> numbers = [4, 2, 4, 5, 2, 3, 5, 1] <i>Output:</i> [4, 2, 5, 3, 1] <pre>def remove_duplicates(lst): unique_list = [] for item in lst: if item not in unique_list: unique_list.append(item) return unique_list # Example input numbers = [4, 2, 4, 5, 2, 3, 5, 1] # Remove duplicates result = remove_duplicates(numbers) print(result) # Output: [4, 2, 5, 3, 1]</pre>
6)	Given a list of numbers and a target sum, find all unique pairs that add up to that sum. And upload add the code to Git repository <i>Example:-</i> Input:- List → [2, 4, 3, 5, 7, 8, -1] Target sum → 6

Output:- [(2, 4), (3, 3), (7, -1)]

```
def find_pairs(lst, target):  
    pairs = []  
    seen = set()  
  
    for num in lst:  
        complement = target - num  
        if complement in lst and (min(num, complement), max(num, complement)) not in  
seen:  
            # Avoid duplicates and count each pair only once  
            pairs.append((num, complement))  
            seen.add((min(num, complement), max(num, complement)))  
    return pairs  
  
# Example input  
numbers = [2, 4, 3, 5, 7, 8, -1]  
target = 6  
  
result = find_pairs(numbers, target)  
print(result) # Output: [(2, 4), (3, 3), (7, -1)]
```